Chiziqli ifodalarni soddalashtirish

1-QISM

Qiziqtiramiz

BUVIJONNING RASTASI

Bozorda buvijon xaridorlar uchun mevalarni likopchalarga ajratib qo'ydi. Hamma olmalar bir xil kattalikda, noklar ham, apelsinlar ham.



3 ta olma | 2 ta nok | 4 ta olma | 5 ta apelsin | 1 ta nok

1-savol. Kechqurun buvijon mevalarni turiga qarab savatlarga qaytarib soldi. Har bir savatda nechta meva bo'ladi?

2-savol. Bitta olmaning massasi a gramm, bitta nokniki n gramm, bitta apelsinniki p gramm. Likopchalardagi barcha mevalar massasini ifoda bilan yozing. Shu ifodani qisqaroq yozib bo'ladimi?



Beshta likopchani sanash oson. Likopcha ellikta bo'lsa-chi? Uzun ifodani qisqa yozishning yo'li bo'lsa kerak.

ESLAB OLAYLIK

- Ifodadagi harf — o'zgaruvchi: $3a$ — bu $3 \cdot a$, a esa $1 \cdot a$.
- Butun sonlarni qo'shish va ayirish: $-7 + 3 = -4$, $-2 - 5 = -7$, $4 - (-6) = 10$.
- Kasrlarni qo'shish: $\frac{1}{4} + \frac{2}{3} = \frac{3}{12} + \frac{8}{12} = \frac{11}{12}$ (EKUK(4; 3) = 12).
- Daraja: $a^2 = a \cdot a$, $a^3 = a \cdot a \cdot a$.

2-QISM

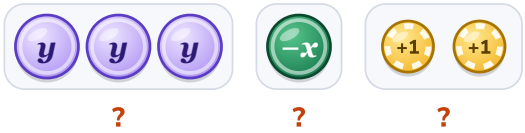
Sinab ko'ramiz

1-TOPSHIRIQ · HARFLI TUGMALAR

Sonlarni sariq (+1) va ko'k (−1) tugmalar bilan tasvirlaganmiz. Endi harfli tugmalar ham bor: yashil tugmaning bir tomonida x, ikkinchi tomonida $-x$; binafsha tugmada y va $-y$.



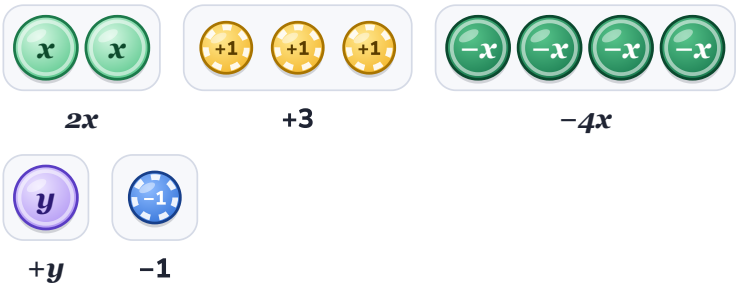
a) Quyidagi tugmalar qaysi ifodani tasvirlaydi?



b) $2x - 3y + 1$ ni tasvirlash uchun qaysi tugmalardan nechta olasiz?
c) $1 + (-1) = 0$ — bunday ikki tugma nol juftlik edi. $x + (-x)$ nimaga teng? 4 ta x va 3 ta $-x$ tugmadan juftliklarni olib tashlasak, nima qoladi?

2-TOPSHIRIQ · TUGMALARNI SARALAYMIZ

Stol ustida $2x + 3 - 4x + y - 1$ ifodaning tugmalari turibdi.



Bir xil rangdagi tugmalarni bitta qatorga yig'ing va nol juftliklarni olib tashlang. Qaysi tugmalar qoldi? Ularni ifoda bilan yozing.

3-TOPSHIRIQ · BIR OILA

Kartochkalar: $5m$, $-2n$, 7, m, $3mn$, -4, $\frac{1}{2}n$, m^2, nm.

Tugmalar kabi birlashtirsa bo'ladiganlarini guruhlarga ajrating. Qaysi kartochka yolg'iz qoladi?

4-TOPSHIRIQ · HISOBLAB SOLISHTIRAMIZ

x	$3x+5x$	$8x$
2	?	?
−1	?	?
10	?	?

Jadvalni to'ldiring. Endi $x=1$, $y=2$ bo'lsa, $3x+5y$ va $8xy$ ni hisoblang. Nimani payqadingiz?

3-QISM

Tushunamiz

$5a-2b+a+3ab-8$ ifodani qaraymiz. Undagi $5a$, $-2b$, a, $3ab$ va -8 — qo'shiluvchilar.

TA'RIF · HAD, OZOD HAD, KOEFFITSIYENT

Algebraik ifodadagi qo'shiluvchilar algebraik ifodaning **hadlari** deyiladi. Algebraik ifodaning hadi son va bir nechta o'zgaruvchilar ko'paytmasidan iborat bo'lishi mumkin. O'zgaruvchi qatnashmagan had **ozod had** deyiladi.

Algebraik ifodaning hadi oldidagi son (ishorasi bilan birga) uning **koeffitsiyenti** deyiladi.

1-MISOL · HADLAR VA OZOD HAD

$5a-2b+a+3ab-8$ da 5 ta had bor: $5a$, $-2b$, a, $3ab$, -8.

Ozod had: -8 — unda harf yo'q.

2-MISOL · KOEFFITSIYENTLAR

$5a \to 5$; $-2b \to -2$; $a \to 1$ (chunki $a = 1\cdot a$);
$3ab \to 3$; $-8 \to -8$.
Yana: $-y \to -1$; $\frac{2}{3}t \to \frac{2}{3}$.

TA'RIF · O'XSHASH HADLAR

Algebraik ifodadagi bir xil o'zgaruvchiga ega hadlar **o'xshash hadlar** deyiladi. Barcha ozod hadlar o'xshash hadlar bo'ladi.

O'xshash hadlarning o'zgaruvchilari bir xil bo'lib, bu o'zgaruvchilarning darajalari ham bir xil bo'ladi.

3-MISOL · O'XSHASH

$4p$ va $-\frac{1}{3}p$ — o'zgaruvchisi p.
$2{,}5mn$ va $-nm$ — o'zgaruvchilari m va n ($nm = mn$).
9 va $-\frac{3}{4}$ — ikkalasi ozod had.

4-MISOL · O'XSHASH EMAS

$4p$ va $4q$ — harflari har xil.
$3p$ va $3p^2$ — darajalari har xil.
pq va p: birinchisida yana q bor.

TA'RIF · HADNING DARAJASI VA CHIZIQLI IFODA

Algebraik ifoda **hadining darajasi** deb bu had o'zgaruvchilari darajalarining yig'indisiga aytiladi. Ozod hadning darajasi 0 ga teng bo'ladi.

Ozod hadidan boshqa barcha hadlari darajasi 1 ga teng algebraik ifoda **chiziqli ifoda** deyiladi.

5-MISOL · HADNING DARAJASI

$7a \to 1$; $-2a^2b \to 2+1=3$;
$mn \to 1+1=2$; $5 \to 0$ (ozod had).

6-MISOL · CHIZIQLIMI?

$4m-3n+1$ — chiziqli: $4m$ va $-3n$ ning darajasi 1.
m^2+2m — chiziqli emas: m^2 ning darajasi 2.
$3mn-1$ — chiziqli emas: $3mn$ ning darajasi 2.

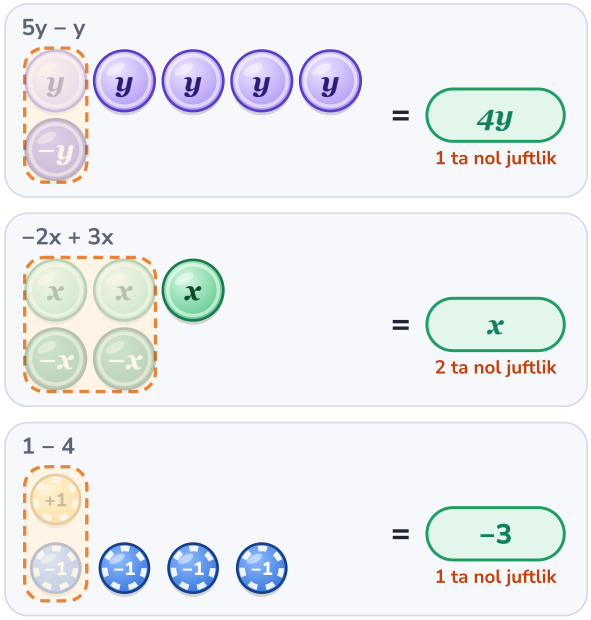
USUL · TUGMALAR BILAN SODDALASHTIRISH

Koeffitsiyentlari butun sonlar bo'lgan chiziqli ifodalarni algebraik tugmalar bilan tasvirlash mumkin. x va $-x$, y va $-y$ ham nol juftlik: $x + (-x) = 0$.

1. Har bir hadni tugmalar bilan tasvirlaymiz.
2. Bir xil rangdagi (o'xshash hadlarning) tugmalarini bitta qatorga yig'amiz.
3. Nol juftliklarni olib tashlaymiz.
4. Qolgan tugmalarni sanab, ifoda qilib yozamiz.

7-MISOL · TUGMALAR BILAN

Soddalashtiring: $5y - 2x + 1 - y + 3x - 4$.



5y − 2x + 1 − y + 3x − 4 = **4y + x − 3**

QOIDA · O'XSHASH HADLARNI IXCHAMLASH

O'xshash hadlarni ixchamlash orqali chiziqli ifodalarni soddalashtirishimiz mumkin. O'xshash hadlarni ham haqiqiy sonlar kabi qo'shish va ayirishimiz mumkin: koeffitsiyentlar qo'shiladi, harfiy qismi o'zgarmaydi.

8-MISOL · BUTUN KOEFFITSIYENTLAR

$9m - 4m + m = (9 - 4 + 1)m = 6m$

$-3d - 5d = (-3 - 5)d = -8d$

9-MISOL · KASR KOEFFITSIYENTLAR

$$\frac{3}{4}k - \frac{1}{6}k = \left(\frac{9}{12} - \frac{2}{12}\right)k = \frac{7}{12}k$$

(EKUK(4; 6) = 12)

$$-2t - \frac{5}{3}t = \left(-\frac{6}{3} - \frac{5}{3}\right)t = -\frac{11}{3}t$$

E'TIBOR BERING

- Agar hadning koeffitsiyenti noto'g'ri kasr bo'lsa, uni aralash son ko'rinishida yozish shart emas: $-\frac{11}{3}t$ deb qoldiramiz.
- Ifoda barcha hadlarning yig'indisidir. O'xshash hadlarni ixchamlashda hadning oldidagi ishorasi saqlanib qolishi kerak: $-6a + 2a = -4a$. Bu yig'indi $-(6a + 2a)$ ham, $(6a + 2a)$ ham emas.
- **Teng kuchlilik.** Soddalashtirishning har bir bosqichida hadlar o'zgarishi mumkin, ammo bunda o'zgaruvchi o'rniga son qiymatini qo'yganda ifodaning qiymati o'zgarmaydi (4-topshiriqdagi jadval).

1-QISMDAGI SAVOLLAR



7 ta olma
$7a$

3 ta nok
$3n$

5 ta apelsin
$5p$

1-savol. Olmalar: $3 + 4 = 7$ ta, noklar: $2 + 1 = 3$ ta, apelsinlar: 5 ta.
2-savol. $3a + 2n + 4a + 5p + n$. $3a$ va $4a$, $2n$ va n — o'xshash hadlar: $3a + 4a = 7a$, $2n + n = 3n$. Qisqasi: $7a + 3n + 5p$ gramm.

4-QISM

Qo'llaymiz

1-MISOL · HADLAR, OZOD HAD, KOEFFITSIYENTLAR

$-k + 6km - \frac{2}{5}m + 11$ ifodada: a) nechta had bor? b) ozod hadi qaysi? c) har bir hadning koeffitsiyenti nechaga teng?

a) Hadlar ishorasi bilan ajratiladi: $-k$, $6km$, $-\frac{2}{5}m$, 11. Jami 4 ta had.

b) Harfsiz had — 11.

c) $-k$ da son yozilmagan, lekin minus bor: -1. Qolganlari: 6, $-\frac{2}{5}$, 11.

2-MISOL · O'XSHASH HADLARNI TOPAMIZ

a) $3b^2 - 5b + \frac{1}{4}b^2$ b) $0{,}6xy + 2x - xy + 4yx$ c) $-5 + c + c^3 + \frac{2}{7}$

a $3b^2$ va $\frac{1}{4}b^2$ — ikkalasida b^2. $-5b$ ularga o'xshash emas: daraja 1.

b $0{,}6xy$, $-xy$ va $4yx$ — hammasida x va y ($yx = xy$). $2x$ — o'xshash emas.

c -5 va $\frac{2}{7}$ — ozod hadlar. c va c^3 — darajalari har xil, o'xshash emas.

3-MISOL · IKKI O'ZGARUVCHI

Soddalashtiring: $7p - 3q - 9p + 8q$.

1 O'xshash hadlarni yonma-yon yozamiz, har had o'z ishorasi bilan ko'chadi: $7p - 9p - 3q + 8q$.

2 Koeffitsiyentlarni qo'shamiz: $(7 - 9)p + (-3 + 8)q$.

3 $= -2p + 5q$. p va q — o'xshash emas, bundan ortiq soddalashmaydi.

4-MISOL · MANFIY SON AYIRILSA

Soddalashtiring: $4a - (-6b) + (-9a) - 2b$.

1 Avval ishoralarni soddalashtiramiz: $-(-6b) = +6b$, $+(-9a) = -9a$. Ifoda: $4a + 6b - 9a - 2b$.

2 $(4 - 9)a + (6 - 2)b = -5a + 4b$.

5-MISOL · KASRLAR VA OZOD HADLAR

Soddalashtiring: $\frac{2}{3}x - 4y + \frac{1}{2} + \frac{1}{6}x + 7y - \frac{3}{4}$.

1 Uch guruh: x li hadlar, y li hadlar va ozod hadlar: $\frac{2}{3}x + \frac{1}{6}x - 4y + 7y + \frac{1}{2} - \frac{3}{4}$.

2 x: $\frac{4}{6} + \frac{1}{6} = \frac{5}{6}$ (EKUK(3; 6) = 6). y: $-4 + 7 = 3$. Ozod: $\frac{2}{4} - \frac{3}{4} = -\frac{1}{4}$.

3 $= \frac{5}{6}x + 3y - \frac{1}{4}$.

6-MISOL · HADLAR KO'PAYTMA KO'RINISHIDA

Soddalashtiring: a) $2a \cdot (-5b) + 3b \cdot 4a$ b) $5m^2n - 2mn + 3nm - m^2n$.

a) Har hadni sonlar va harflarga ajratib ko'paytiramiz: $2a \cdot (-5b) = -10ab$, $3b \cdot 4a = 12ba = 12ab$. Demak, $-10ab + 12ab = 2ab$.

b) $5m^2n$ va $-m^2n$ — o'xshash; $-2mn$ va $3nm$ — o'xshash. $(5-1)m^2n + (-2+3)mn = 4m^2n + mn$.

7-MISOL · SODDALASHTIRIB, QIYMATINI TOPAMIZ

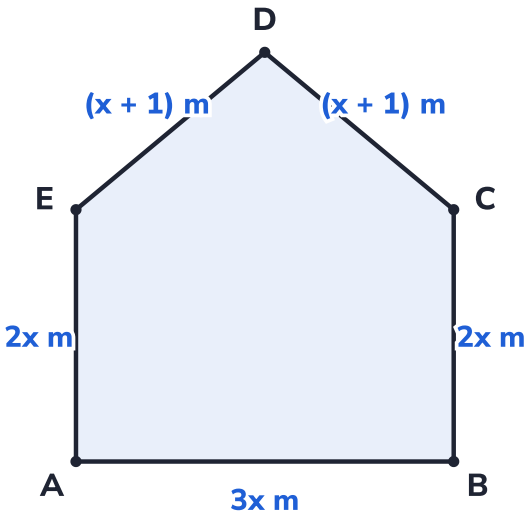
a) Soddalashtiring: $3 - 5c + 4d + 2c - 7 - d$. b) $c = -2$, $d = 3$ bo'lsa, qiymatini toping.

a) $(-5+2)c + (4-1)d + (3-7) = -3c + 3d - 4$.

b) $-3 \cdot (-2) + 3 \cdot 3 - 4 = 6 + 9 - 4 = 11$. Soddalashtirilgan ifodaga qo'yish ancha oson.

8-MISOL · GULZORNING PERIMETRI

Uycha shaklidagi gulzor atrofiga to'siq qo'yiladi. Tomonlari rasmda berilgan.



a) To'siq uzunligini x orqali ifodalang. b) $x = 4$ bo'lsa, to'siq necha metr?

a) Perimetr — barcha tomonlar yig'indisi: $3x + 2x + x + 1 + x + 1 + 2x$. x li hadlar: $3 + 2 + 1 + 1 + 2 = 9$; ozod hadlar: $1 + 1 = 2$. To'siq: $(9x + 2)$ m.

b) $9 \cdot 4 + 2 = 38$. To'siq 38 m.

9-MISOL · VELOMARAFON

Velomarafonning 1-kunida t km yo'l bosildi. 2-kunida 1-kundagidan 15 km ko'p, 3-kunida esa birinchi ikki kunda birgalikda bosilgan yo'ldan 20 km kam yo'l bosildi. a) Uch kundagi jami yo'lni t orqali ifodalang. b) $t = 48$ bo'lsa, jami necha km?

1. 1-kun: t; 2-kun: $t + 15$; 3-kun: $t + t + 15 - 20 = 2t - 5$.
2. Jami: $t + t + 15 + 2t - 5 = (1 + 1 + 2)t + (15 - 5) = 4t + 10$ km.
3. $t = 48$: $4 \cdot 48 + 10 = 202$. Jami 202 km.



Hadni ko'chirganda uning oldidagi ishorani ham birga olaman. $-7c$ dagi minus c bilan birga ketadi.

DIQQAT!

1. Ishora saqlanadi: ✗ $-7c + 3c = -10c$ ✓ $-7c + 3c = (-7 + 3)c = -4c$
2. Harflari har xil hadlar qo'shilmaydi: ✗ $5a + 2b = 7ab$ ✓ $5a + 2b$ — **soddalashmaydi**
3. Darajalari har xil hadlar o'xshash emas: ✗ $4m^2 + 3m = 7m^3$ ✓ $4m^2 + 3m$ — **soddalashmaydi**
4. Yozilmagan koeffitsiyent — 1: ✗ $6k - k = 6$ ✓ $6k - k = (6 - 1)k = 5k$
5. Ozod had harfli hadga qo'shilmaydi: ✗ $3x + 2 = 5x$ ✓ $3x + 2$ — **soddalashmaydi**

Mashqlar

1. Hadlar soni va ozod hadni toping.

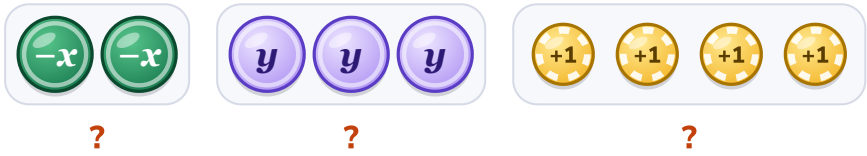
   a) $4m - 7n + 2$ b) $-6 + 3a - b + 5c$ c) $\frac{2}{9}t^2 - t + 0{,}8$ d) $xy - 1$

2. x va y ning koeffitsiyentlarini yozing.
   a) $2xy - 7y + x$ b) $-x + \frac{3}{8}y - 4$ c) $0{,}7y - 9 - 2{,}3x$ d) $y^2 - 4x + 3y$

3. Juftlikdagi hadlar o'xshashmi? Javobingizni tushuntiring.
   a) $-3xy^2$ va $8x^2y$ b) $\frac{2}{5}ab$ va $-ba$ c) $4k^3$ va $4k^2$
   d) $1{,}2mnp$ va $-pnm$ e) $2^3 \cdot c$ va $c \cdot 3$ f) -6 va 2^5 g) $10y$ va $10yz$

4. Har bir hadning darajasini aniqlang. Ifoda chiziqlimi?
   a) $3a - 2b + 7$ b) $m^2 - 4m$ c) $\frac{1}{2}x + 0{,}4y - z$ d) $5pq + p$ e) $-6t^3 + 1$

5. a) Tugmalar qaysi ifodani tasvirlaydi?



   b) $2y - 3 - 4y + 5$ ni tugmalar bilan soddalashtiring: nechta nol juftlik olib tashlanadi, nima qoladi?

6. Soddalashtiring.
   a) $8a + 5a$ b) $4b - 9b$ c) $-7c + 10c$ d) $-3d - 6d$
   e) $m - 6m + 3m$ f) $-5n + 2n + 3n$ g) $4p - p - 8p$ h) $-k - 2k - 5k + 9k$

7. Soddalashtiring.
   a) $4 - 3a + 9 + 5a - 2$ b) $-8 + t - 6t + 3t$ c) $-(-2x) + 6y - 9x - (-y)$
   d) $5n + (-3m) - (-4m) - 8n + m$ e) $3c - 7d + 4 - 5c + 9d - 10$

8. Soddalashtiring.
   a) $\frac{1}{3}a + \frac{1}{2}a$ b) $\frac{5}{6}b - \frac{3}{4}b$ c) $2c - \frac{7}{5}c$
   d) $-\frac{1}{2}y + \frac{3}{8}x + \frac{5}{6}y - x$ e) $-\frac{2}{3} + 0{,}5m - \frac{1}{6} + m$

9. Soddalashtiring.
   a) $5y \cdot (-2x) - 3x \cdot (-6y)$ b) $7ab - 2a + 3ba + a$
   c) $2m^2n + mn - 5nm^2 + 4nm$ d) $\frac{5}{2}pq - 3pq + \frac{1}{4}qp$

10. a) Soddalashtiring: $9 - 4x + 3 + 6x$. b) $x = -7$ bo'lsa, ifodaning qiymatini toping.

11. a) Soddalashtiring: $3a - (-2b) + 4ab - 5a - (-ab) - 7b$. b) $a = 2, b = -1$ bo'lsa, qiymatini toping.

12. a) Soddalashtiring: $\frac{2}{3}m + \frac{1}{4} + 2n - \frac{1}{6}m - 3n + \frac{1}{2}$. b) $m = -6, n = -2$ bo'lsa, qiymatini toping.

13. **Xatoni toping.** Nodira shunday yechdi: $6a - 4 - 9a + 2a^2 = 3a - 4 + 2a^2 = 5a^3 - 4$. U qayerda adashdi? To'g'ri javobni yozing.

14. Gugurt cho'plaridan bir qatorda n ta kvadrat terildi. Tepada n ta, pastda n ta, tik turgan $(n + 1)$ ta cho'p bor.
    a) Cho'plar soni $n + n + n + 1$ ni soddalashtiring. b) 25 ta kvadrat uchun nechta cho'p kerak?

15. Ertalab harorat t °C edi. Tushgacha harorat 7 °C ga ko'tarildi, kechqurun 10 °C ga, tunda yana 4 °C ga pasaydi.
    a) Tundagi haroratni t orqali ifodalang va soddalashtiring. b) $t = -3$ bo'lsa, tunda harorat necha gradus bo'lgan?

16. To'g'ri to'rtburchak shaklidagi rasm ramkasining bo'yi $(2a + 3)$ cm, eni $(a - 1)$ cm.
    a) Ramka perimetrini to'rt tomon yig'indisi sifatida yozib, soddalashtiring. b) $a = 10$ bo'lsa, perimetr necha cm?

17. Uch kun yomg'ir yog'di: 1-kuni $(2a + 5)$ mm, 2-kuni $(3a - 1)$ mm, 3-kuni $(a + 2)$ mm.
    a) Uch kundagi jami yog'inni soddalashtirilgan ifoda bilan yozing.
    b) $a = 4$ bo'lsa, qaysi kuni eng ko'p yog'gan? Jami va bir kunda o'rtacha necha mm yog'gan?

18. Ifodaga mos masala tuzing, so'ng ifodani soddalashtiring: $x + 2x + (x - 4)$. Bunda x — biror narsaning soni yoki narxi.

5-QISM

Tekshiramiz

1. $-3p + 8 - q + 2pq$ ifodada hadlar sonini, ozod hadni va har bir hadning koeffitsiyentini toping.

2. O'xshash hadlarmi? a) $5x^2y$ va $-yx^2$ b) $9ab$ va $9a$ c) -4 va $\frac{1}{3}$

3. Qaysi ifoda chiziqli? a) $2m + 3n - 1$ b) $m^2 - n$ c) $4mn + 5$

4. Soddalashtiring.
   a) $11a - 4a$ b) $-2b - 7b + 5b$ c) $5c - 3 - 8c + 9$ d) $-(-5y) + 4x - 7x - 8y$

5. Soddalashtiring. a) $\frac{2}{5}m + \frac{1}{2}m$ b) $\frac{3}{4}k - 2t - \frac{1}{3}k + \frac{1}{2}t$

6. Soddalashtiring: $3a \cdot (-2b) + 4b \cdot a$.

7. a) Soddalashtiring: $8 - 2y + 5y - 11$. b) $y = -3$ bo'lsa, qiymatini toping.

8. Rasmdagi tugmalar ifodasini yozing va soddalashtiring.



9. Uchburchak shaklidagi gulzorning tomonlari x m, $(x + 4)$ m va $(2x - 1)$ m. a) Perimetrni x orqali ifodalang. b) $x = 6$ bo'lsa, perimetr necha metr?

10. **Xatoni toping.**
    a) Olim: «$7m - 2 - 3m + 5 = 10m + 3$.»
    b) Laylo: «$2a + 3b = 5ab$.»
    c) Bobur: «$x^2 + x^2 = x^4$.»

- [ ] Men ifodaning hadlarini, ozod hadini va koeffitsiyentlarini topa olaman.
- [ ] Men o'xshash hadlarni taniyman: harflari va darajalari bir xil.
- [ ] Men ifoda chiziqli yoki chiziqli emasligini hadlarning darajasi bo'yicha aniqlay olaman.
- [ ] Men o'xshash hadlarni ixchamlab, ifodani soddalashtira va qiymatini topa olaman.

Javoblar

4-qism · Mashqlar

1-masala

a) 3 ta had, ozod had 2. b) 4 ta had, ozod had −6. c) 3 ta had, ozod had 0,8. d) 2 ta had, ozod had −1.

2-masala

a) x: 1, y: −7 ($2xy$ — alohida had). b) x: −1, y: $\frac{3}{8}$. c) x: −2,3, y: 0,7. d) x: −4, y: 3 (y^2 — alohida had).

3-masala

a) Yo'q — xy^2 da y ning darajasi 2, x^2y da x ning darajasi 2. b) Ha — $ba = ab$. c) Yo'q — darajalari 3 va 2. d) Ha — $pnm = mnp$. e) Ha — $8c$ va $3c$. f) Ha — ikkalasi ozod had. g) Yo'q — ikkinchisida z ham bor.

4-masala

a) 1, 1, 0 — chiziqli. b) 2, 1 — chiziqli emas. c) 1, 1, 1 — chiziqli. d) 2, 1 — chiziqli emas. e) 3, 0 — chiziqli emas.

5-masala

a) $-2x + 3y + 4$.

b)



y qatorida 2 ta, sonlar qatorida 3 ta nol juftlik. Qoladi: 2 ta $-y$ va 2 ta $+1$: $-2y + 2$.

6-masala

a) $13a$. b) $-5b$. c) $3c$. d) $-9d$. e) $-2m$. f) 0. g) $-5p$. h) k.

7-masala

a) $2a + 11$. b) $-2t - 8$. c) $2x + 6y - 9x + y = -7x + 7y$. d) $5n - 3m + 4m - 8n + m = 2m - 3n$. e) $-2c + 2d - 6$.

8-masala

a) $\frac{2}{6} + \frac{3}{6} = \frac{5}{6} : \frac{5}{6} a$. b) $\frac{10}{12} - \frac{9}{12} : \frac{1}{12} b$. c) $\frac{10}{5} - \frac{7}{5} : \frac{3}{5} c$. d) $\frac{3}{8} - \frac{8}{8} = -\frac{5}{8}, -\frac{3}{6} + \frac{5}{6} = \frac{2}{6} = \frac{1}{3} : -\frac{5}{8} x + \frac{1}{3} y$. e) $0{,}5m + m = 1{,}5m, -\frac{4}{6} - \frac{1}{6} = -\frac{5}{6} : 1{,}5m - \frac{5}{6}$.

9-masala

a) $-10xy + 18xy = 8xy$. b) $10ab - a$. c) $-3m^2n + 5mn$. d) $\frac{10}{4} - \frac{12}{4} + \frac{1}{4} = -\frac{1}{4} : -\frac{1}{4} pq$.

10-masala

a) $2x + 12$. b) $2 \cdot (-7) + 12 = -14 + 12 = -2$.

11-masala

a) $3a + 2b + 4ab - 5a + ab - 7b = -2a - 5b + 5ab$. b) $-2 \cdot 2 - 5 \cdot (-1) + 5 \cdot 2 \cdot (-1) = -4 + 5 - 10 = -9$.

12-masala

a) $\frac{4}{6} - \frac{1}{6} = \frac{1}{2}, 2n - 3n = -n, \frac{1}{4} + \frac{2}{4} = \frac{3}{4} : \frac{1}{2} m - n + \frac{3}{4}$. b) $\frac{1}{2} \cdot (-6) - (-2) + \frac{3}{4} = -3 + 2 + \frac{3}{4} = -\frac{1}{4}$.

13-masala

1-xato: $6a - 9a = -3a$, $3a$ emas (ishora). 2-xato: a va a^2 o'xshash emas, ularni qo'shib bo'lmaydi. To'g'risi: $2a^2 - 3a - 4$.

14-masala

a) $3n + 1$. b) $3 \cdot 25 + 1 = 76$ ta cho'p.

15-masala

a) $t + 7 - 10 - 4 = t - 7$. b) $-3 - 7 = -10$ °C.

16-masala

a) $2a + 3 + a - 1 + 2a + 3 + a - 1 = 6a + 4$ cm. b) $6 \cdot 10 + 4 = 64$ cm.

17-masala

a) $2a + 5 + 3a - 1 + a + 2 = 6a + 6$ mm. b) 1-kuni 13 mm, 2-kuni 11 mm, 3-kuni 6 mm — eng ko'p 1-kuni. Jami $6 \cdot 4 + 6 = 30$ mm, o'rtacha $30 : 3 = 10$ mm.

18-masala

Masalan: «Kutubxonaga dushanba kuni x ta kitob keltirildi, seshanba kuni ikki baravar ko'p, chorshanba kuni dushanbadagidan 4 ta kam. Jami nechta kitob keltirildi?» Soddalashtirilgan ifoda: $4x - 4$.

5-qism · Tekshiramiz

1-masala

4 ta had: $-3p$, 8, $-q$, $2pq$. Ozod had 8. Koeffitsiyentlar: −3, 8, −1, 2.

2-masala

a) Ha — $yx^2 = x^2y$. b) Yo'q — ikkinchisida b yo'q. c) Ha — ikkalasi ozod had.

3-masala

Faqat a) chiziqli. b) da m^2, c) da $4mn$ ning darajasi 2.

4-masala

a) $7a$. b) $-4b$. c) $-3c + 6$. d) $5y + 4x - 7x - 8y = -3x - 3y$.

5-masala

a) $\frac{4}{10} + \frac{5}{10} : \frac{9}{10} m$. b) $\frac{9}{12} - \frac{4}{12} = \frac{5}{12}$, $-2 + \frac{1}{2} = -\frac{3}{2} : \frac{5}{12} k - \frac{3}{2} t$.

6-masala

$-6ab + 4ab = -2ab$.

7-masala

a) $3y - 3$. b) $3 \cdot (-3) - 3 = -12$.

8-masala

$3x - 1 - 5x + 4 = -2x + 3$.

9-masala

a) $x + x + 4 + 2x - 1 = 4x + 3$ m. b) $4 \cdot 6 + 3 = 27$ m.

10-masala

a) $-3m$ ning ishorasi unutilgan: $7m - 3m = 4m$, to'g'risi $4m + 3$. b) a va b o'xshash emas — $2a + 3b$ soddalashmaydi. c) Koeffitsiyentlar qo'shiladi, daraja o'zgarmaydi: $x^2 + x^2 = 2x^2$.